

# Few-Step MeanFlow Residuals for Convection-Allowing Forecasting over Taiwan

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日期：

# Outline

- Introduction
- Related work
- Dataset
- Method
- Experiments
- Discussion
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# Introduction

- Hourly regional convection-allowing generative forecasting model (StormCast) is computationally expensive.
  - 35 network calls per forecast steps  $\rightarrow$  a 50-member, 24-h ensemble  $\approx$  42000 U-Net passes.
- Precipitation is hard to predict yet on generative-based forecasting models.

# Motivation

- A sharp hourly regional convection-allowing generative forecasting model
  - Time resolution: 1 hour
  - Spatial resolution: ~2 km
  - Inference at 1-2 steps
  - Achieve similar or better forecasting quality

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# Numerical Weather Prediction

- Solve the physics on a grid (IFS, GFS, WRF/RWRF). Accurate, physically grounded, expensive.

# Resolutions and the CFL condition

- Spatial resolution
  - grid spacing / pixel resolution of weather data
  - Two regimes:
    - Synoptic / Global (~10-25 km): resolves cyclones
    - Convection-allowing / regional (~1-4 km): resolves thunderstorms, orographic precipitation
- Temporal resolution
  - time step between two adjacent weather data
- Changes in resolutions has to obey the Courant–Friedrichs–Lewy condition to be mathematically stable

$$C = \left( \frac{|u|}{\Delta x} + \frac{|v|}{\Delta y} + \frac{|w|}{\Delta z} \right) \Delta t \leq C_{\max} \quad \left( C_{\max} \sim 1 \text{ for explicit schemes} \right) \Delta x \downarrow 2\times \implies \Delta t \downarrow 2\times \implies \text{3-D cost} \uparrow 2^4 = 16\times$$



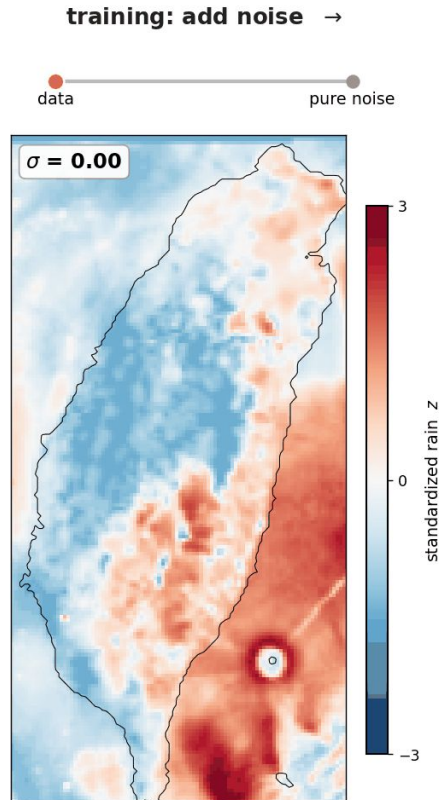
# Numerical Weather Prediction's computational requirement

- ECMWF (European Centre for Medium-Range Weather Forecasts)
  - Global operational weather forecasting and massive climate re-analysis (ERA5)
  - CPU: 1,015,808 AMD EPYC Rome CPU cores
  - Memory: 2,050 Tb
  - 30.0 PetaFLOPS ( $10^{15}$ )
- RIKEN Center for Computational Science
  - "Mega-ensemble" forecasting. (e.g., Running 1,000 simultaneous ensemble members to accurately predict probabilistic events like the July 2020 Kyushu heavy rain)
  - CPU: 7,630,848 Fujitsu A64FX cores
  - Memory: 4,850 Tb
  - 997 PetaFLOPS ( $10^{15}$ )

# The mesoscale bottleneck

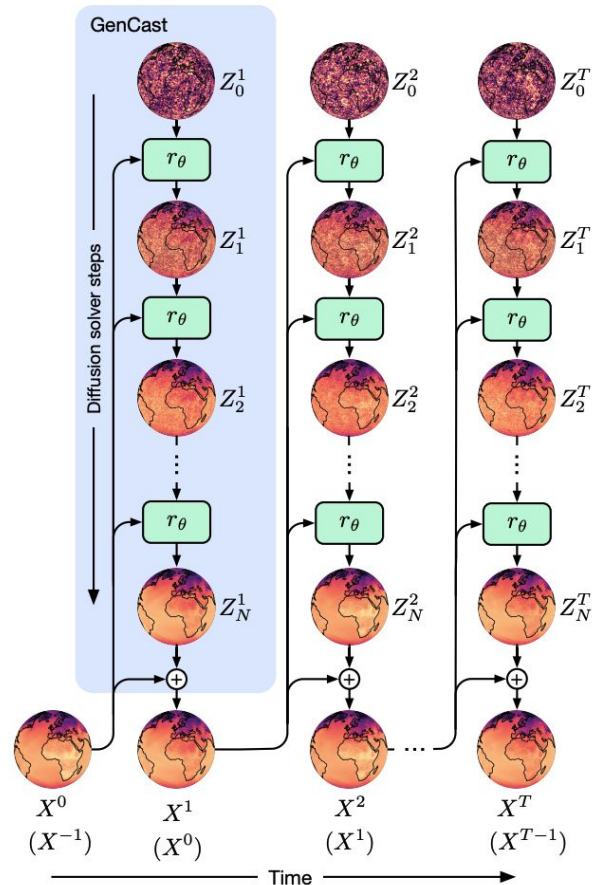
- Synoptic scale ( $\sim 0.25^\circ$  grid, GFS / ECMWF-IFS): cheap, runs every 6 h.
- Convection-allowing models / CAMs ( $\sim 3$  km grid, HRRR / RWRF): must resolve thunderstorms, cold pools, gravity waves
  - $25/3 = 8.333$  times of change in resolution in each dimension
  - To obey CFL, computation is  $8.333^3$  (3 dimensions) \* 8.333 (temporal change) =  $\sim 4821$  more expensive

# Diffusion model

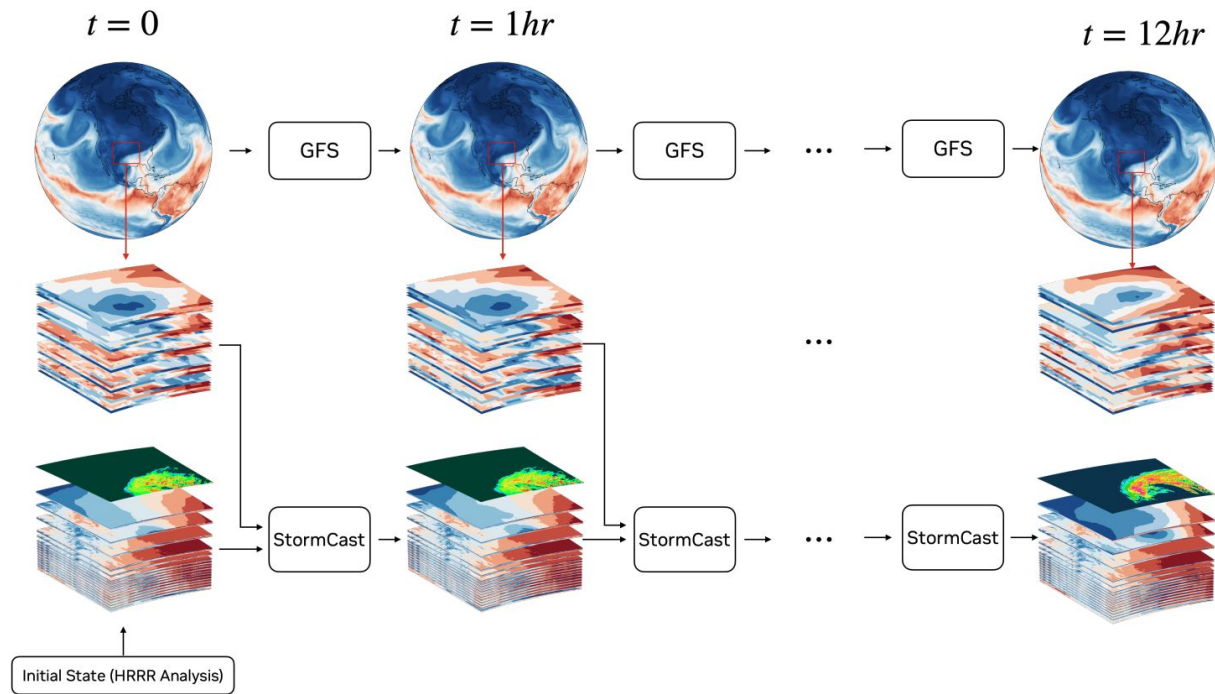


# Google Deepmind's GenCast

- 15-day ensemble
- Spatial resolution:  $0.25^\circ$  (ERA5)
- Time resolution: 12 hours
- Domain: Global



# Nvidia's StormCast

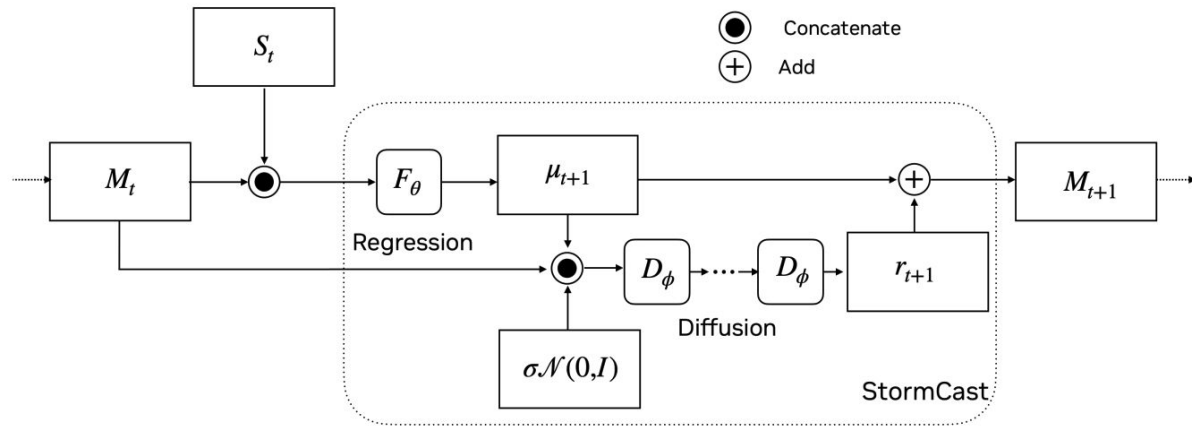


source: [Pathak et al. \(2024\), arXiv:2408.10958](https://arxiv.org/abs/2408.10958)

# Nvidia's StormCast

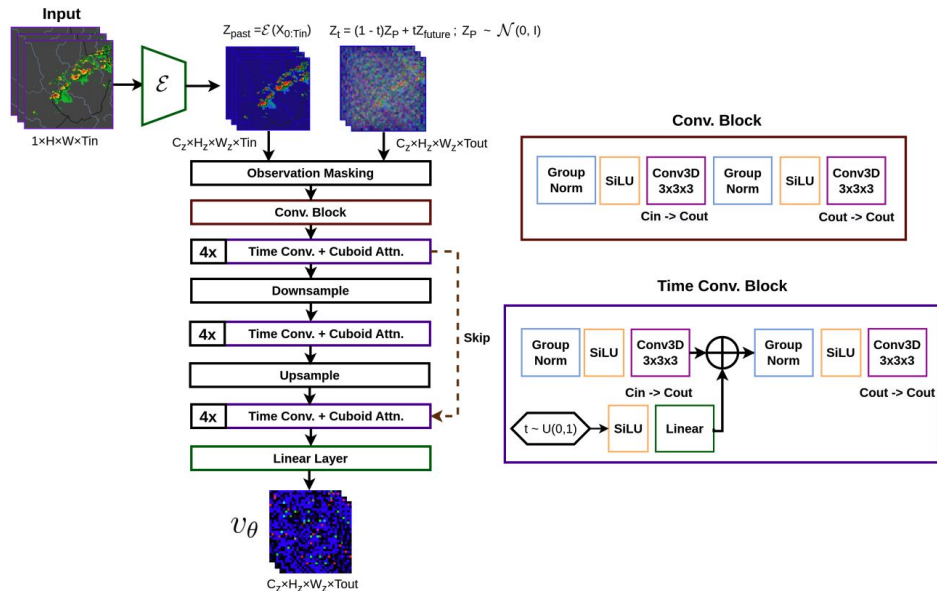
- 6-hour roll out
- 5-member ensemble
- Spatial resolution: 3 km
- Time resolution: 1 hour
- Coverage: Central US
- Domain:

1536 km × 1920 km



# FlowCast

- Non-autoregressive
- Spatial resolution: 1 km
- Time resolution: 5 minutes
- Coverage: US and Solvenian
- Ingests: 13 past frames
- Predicts: 12 futures frames



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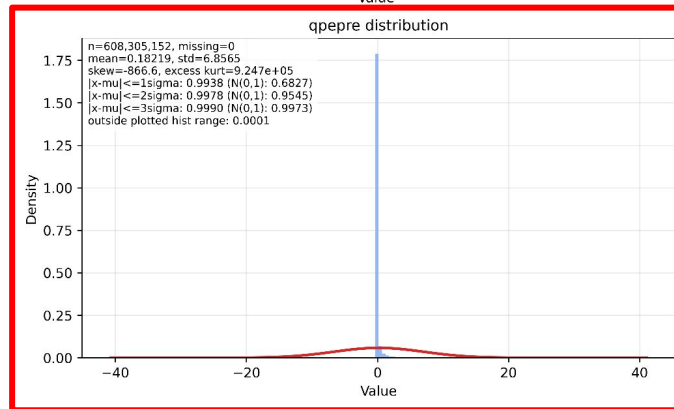
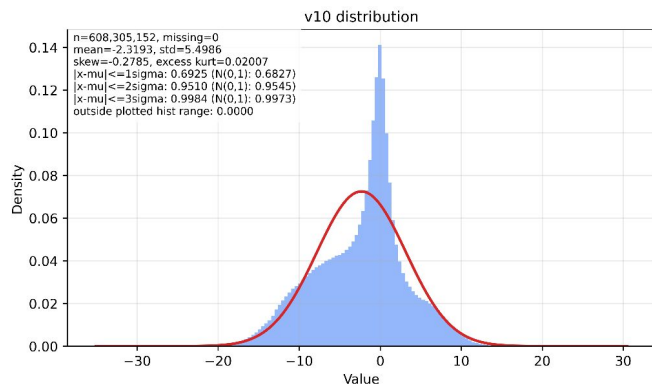
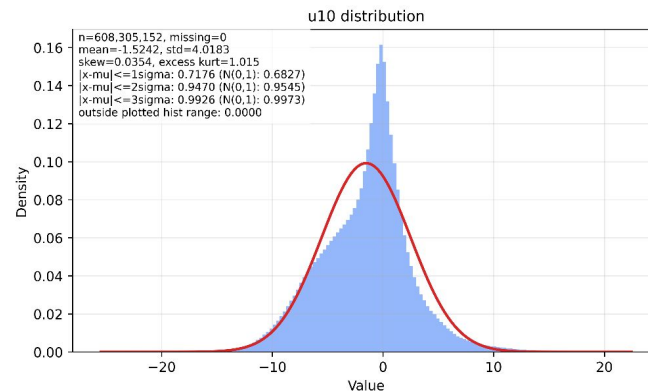
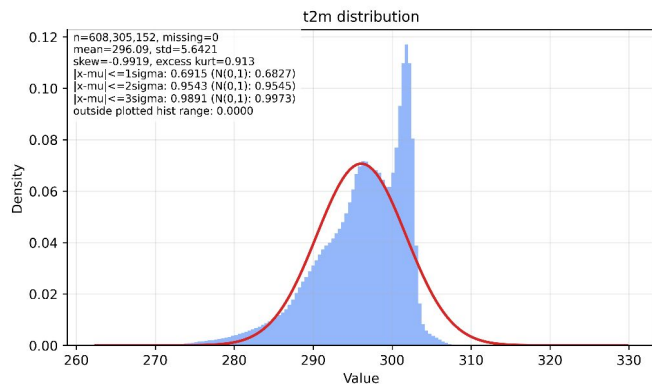
# Dataset: ERA5, RWRF, Qpepre

- Latitude: 21.6° - 25.6° N
- Longitude: 119.75° - 122.25° E
- Time resolution: 1 hour
- Window: 2019/08 to 2022/12
- Samples: 29976 samples

|               | ERA5                             | RWRF                 | QPEPRE               |
|---------------|----------------------------------|----------------------|----------------------|
| Type          | Reanalysis                       | Numerical model      | Radar observation    |
| Resolution    | $\approx 25$ km                  | $\approx 2$ km       | $\approx 2$ km       |
| Role          | LowRes input $S_t$               | HighRes target $X_t$ | HighRes target $X_t$ |
| Channels used | 24 (4 sfc + $5 \times 4$ levels) | 3 (t2m, u10, v10)    | 1 (qpepre)           |
| Native format | NetCDF                           | NetCDF (wrfout)      | ASCII text           |

| Code                                              | Variable                | Units                      | Levels (hPa)        | # ch.     |
|---------------------------------------------------|-------------------------|----------------------------|---------------------|-----------|
| <i>Surface (single level)</i>                     |                         |                            |                     |           |
| mslp                                              | Mean sea-level pressure | Pa                         | surface             | 1         |
| t2m                                               | 2 m temperature         | K                          | surface             | 1         |
| u10                                               | 10 m zonal wind         | $\text{m s}^{-1}$          | surface             | 1         |
| v10                                               | 10 m meridional wind    | $\text{m s}^{-1}$          | surface             | 1         |
| <i>Upper-air (one channel per pressure level)</i> |                         |                            |                     |           |
| q                                                 | Specific humidity       | $\text{kg kg}^{-1}$        | 1000, 850, 500, 250 | 4         |
| t                                                 | Temperature             | K                          | 1000, 850, 500, 250 | 4         |
| u                                                 | Zonal wind              | $\text{m s}^{-1}$          | 1000, 850, 500, 250 | 4         |
| v                                                 | Meridional wind         | $\text{m s}^{-1}$          | 1000, 850, 500, 250 | 4         |
| z                                                 | Geopotential            | $\text{m}^2 \text{s}^{-2}$ | 1000, 850, 500, 250 | 4         |
| <b>Total</b>                                      |                         |                            |                     | <b>24</b> |

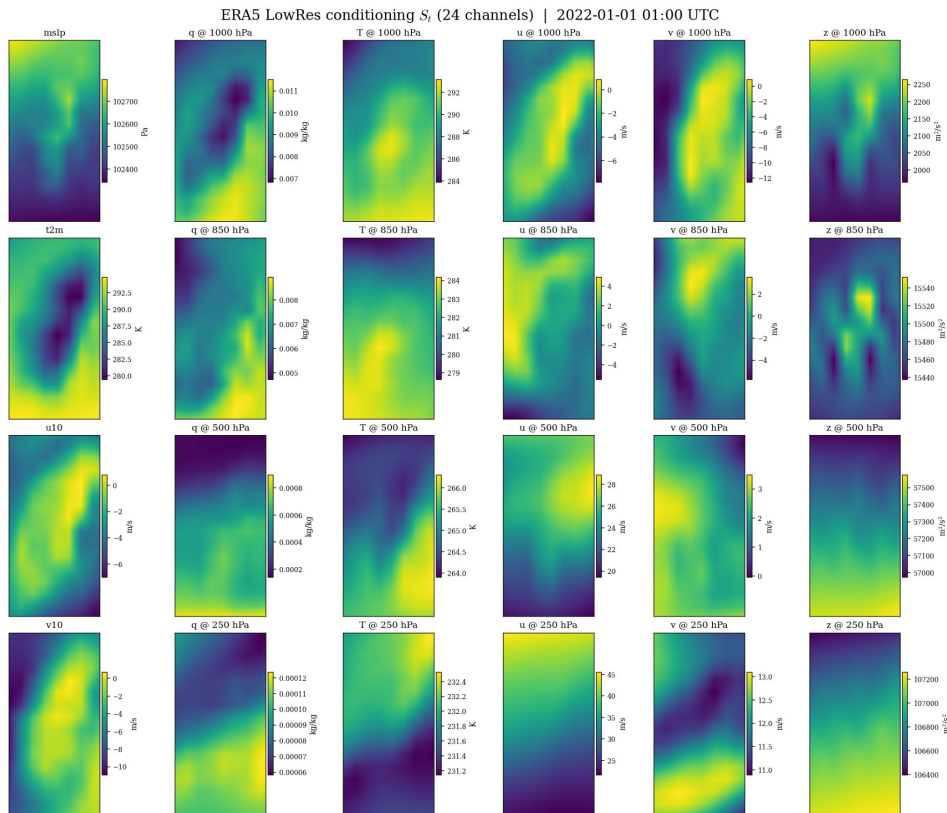
# RWRF & Qpepre distributions



# Log1p on Qpepre distributions

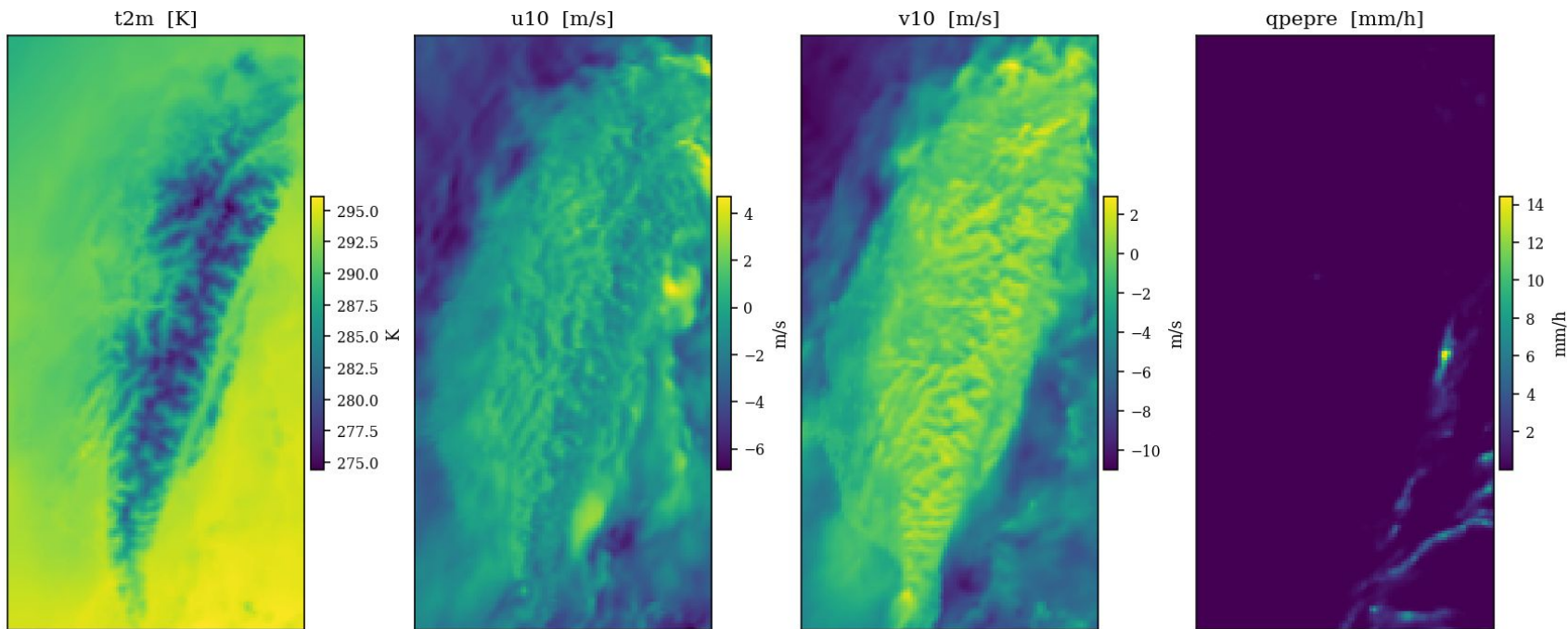
- MSE loss cares about absolute difference
- Predicting 0.1 mm/h instead of 1 mm/h (10× error) contributes the same  $0.81 \text{ mm}^2/\text{h}^2$  to the loss as predicting 100 vs 100.9 mm/h (0.9% error). The model is told both mistakes are equally bad.
- log1p makes the loss measure log-ratios ( $\log_{1p}(1) - \log_{1p}(0.1) \approx 0.60$  vs  $\log_{1p}(100.9) - \log_{1p}(100) \approx 0.009$ ), so the gradient is concentrated where the physics actually differ.
- Perform  $\log_{1p}$ ,  $\ln(1 + x)$  where  $x$  is qpepre before training.
- Inverse with  $e^p - 1$  for output, where  $p$  is transformed qpepre.

# Dataset: ERA5



# Dataset: RWRF and Qpepre

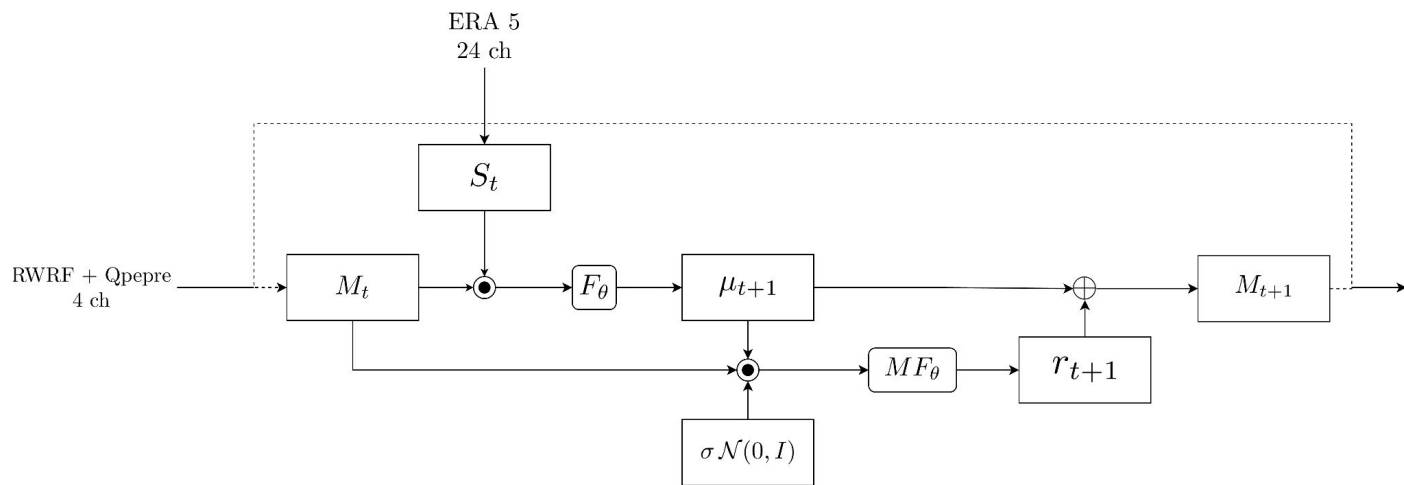
HighRes target  $X_{t+1}$  (4 channels) | 2022-01-01 01:00 UTC



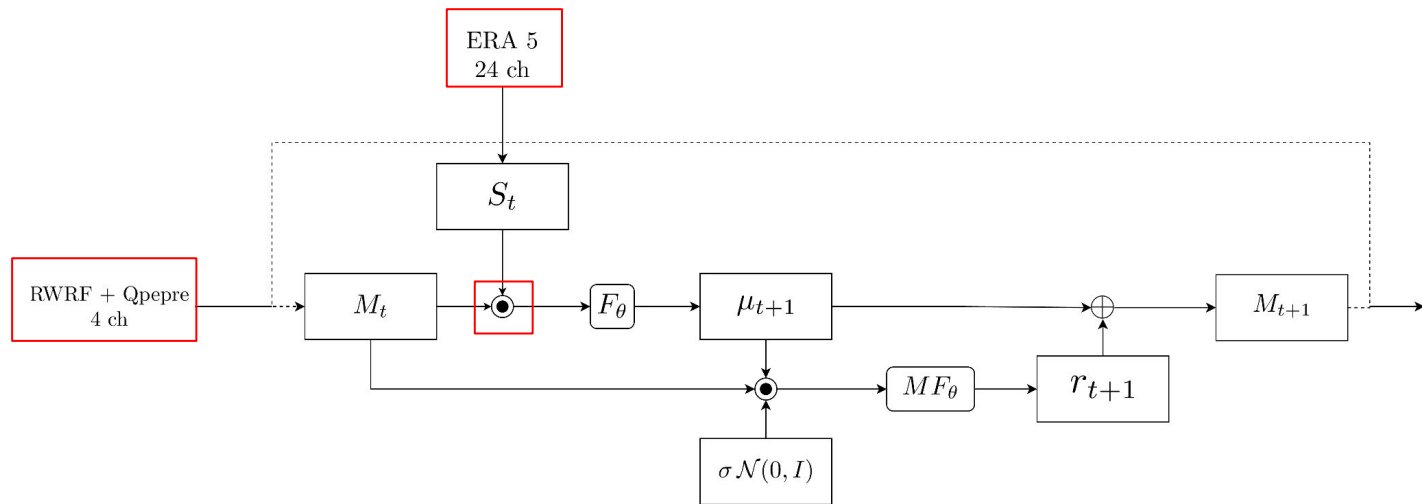
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# Method: Overview



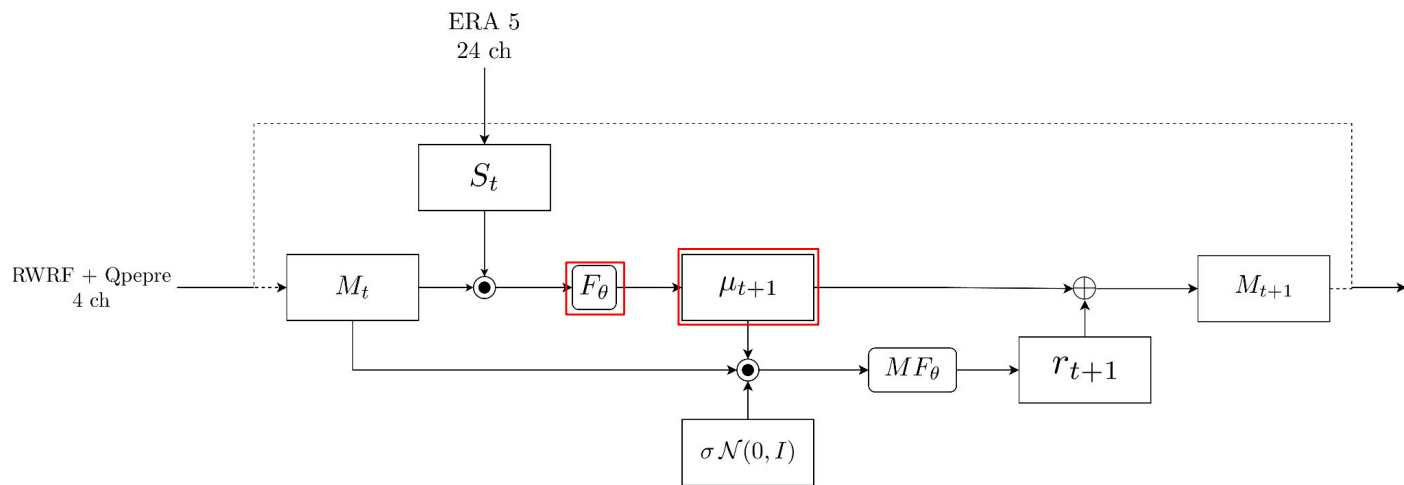
# Method: Input and preprocessing



- Low res and high res data regridded into desired domain
- Concatenated as a tensor



# Method: Regression model

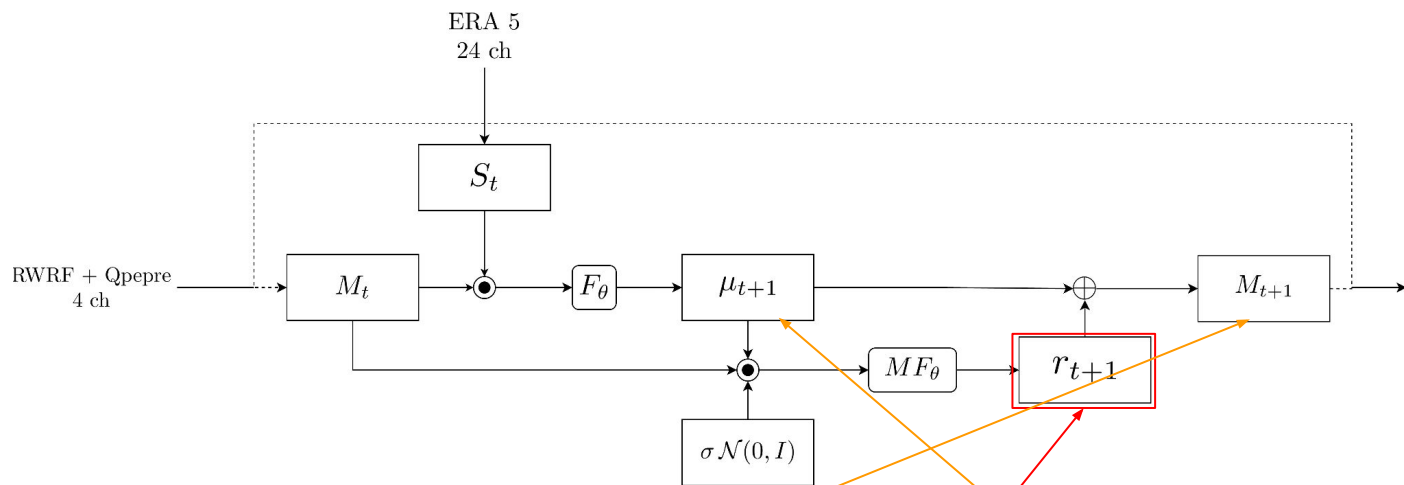


$$F_\theta(M_t, S_t) = \mathbb{E}[M_{t+1} \mid M_t, S_t]$$

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \|M_{t_{i+1}} - F_\theta(M_{t_i}, S_{t_i})\|_2^2$$

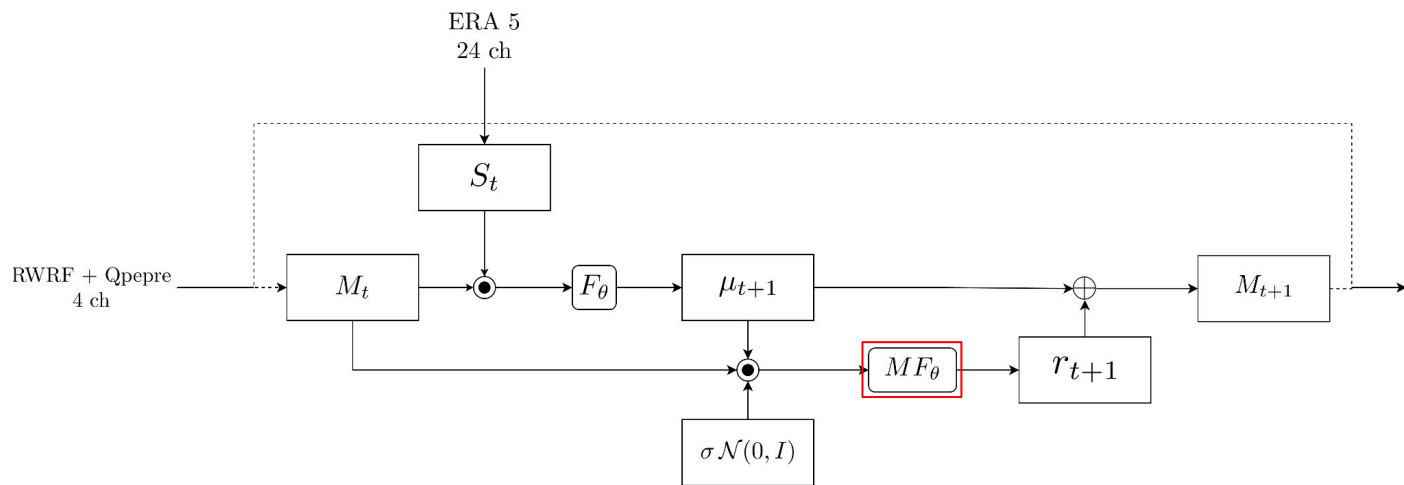
- Train a U-Net learn a deterministic regression  $F_\theta$  to estimate the conditional mean.

# Method: Residual



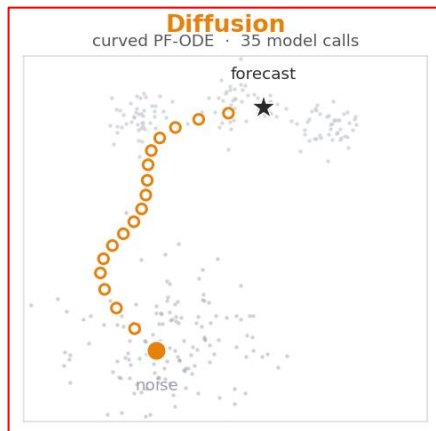
$$p(M_{t+1} | M_t, S_t) = r_{t+1} + \mu_{t+1}$$

# Method: MeanFlow



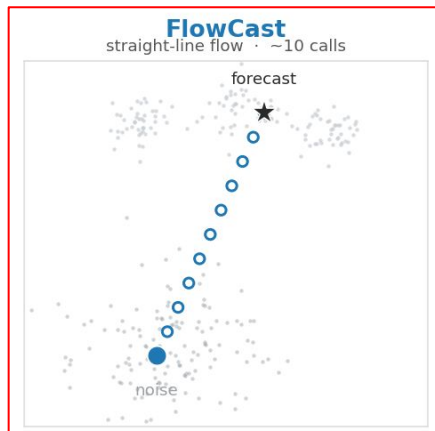
- Train a generative model to learn residual of next timestep

# Method: Diffusion (Original Stormcast)



- StormCast uses EDM, an improved DDPM through better noise scheduling

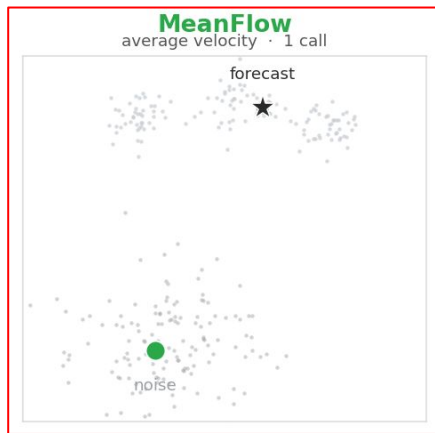
# Method: Conditional FlowMatching (FlowCast)



$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{\substack{t \sim \mathcal{U}(0,1) \\ (x_0, x_1) \sim q \\ x \sim p_t(x | x_0, x_1)}} \left[ \left\| v_{\theta}(t, x) - u_t(x | x_0, x_1) \right\|_2^2 \right]$$

- Conditional flow matching learns the instantaneous velocity at any point between 0 to 1 to go from pure noise to target distribution (data)

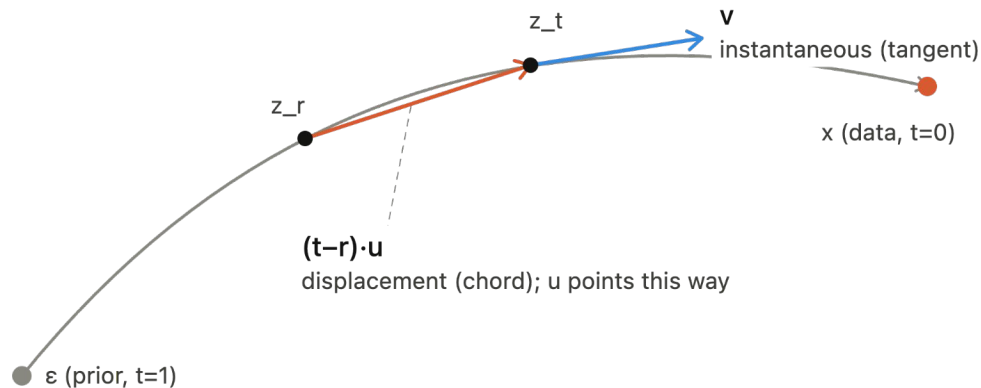
# Method: MeanFlow



$$u(z_r, r, t) \triangleq \frac{1}{t - r} \int_r^t v(z_\tau, \tau) d\tau,$$

- $U$  is the time-average velocity over the interval  $[r, t]$
- $v$  is the instantaneous velocity from FlowMatching, local tangent at each point of the path

# Method: MeanFlow



$$z_t = z_r + (t - r) u(z_r, r, t),$$

- Rearrangement, is  $(r, t)$  is set to  $(0, 1)$  it is a one-step evaluation
- $u$  is still intractable

## Method: MeanFlow Identity

$$u(z_r, r, t) \triangleq \frac{1}{t-r} \int_r^t v(z_\tau, \tau) d\tau,$$



multiply by (t-r) then differentiate w.r.t. to r

$$u(z_r, r, t) = v(z_r, r) + (t-r) \frac{d}{dr} u(z_r, r, t),$$

- Hence, integrate is killed.



## Method: MeanFlow

$$u(z_r, r, t) = v(z_r, r) + (t - r) \frac{d}{dr} u(z_r, r, t),$$

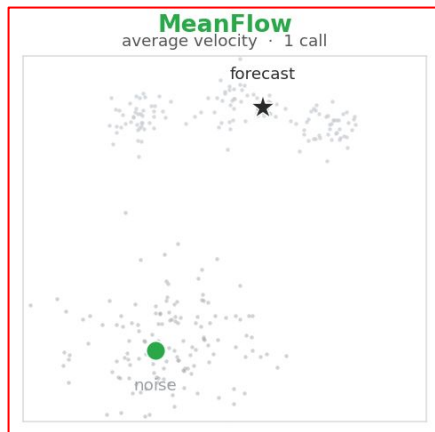


total derivative to r

$$\frac{d}{dr} u(z_r, r, t) = v(z_r, r) \partial_z u + \partial_r u.$$

- The state  $Z_r$  moves along the trajectory as  $r$  changes
- This form maps directly onto a Jacobian-vector product (JVP), one extra forward pass with tangent  $(v, 1, 0)$  for  $(z, r, t)$ , no higher-order gradients.
- In short, the identity is now **trainable**.

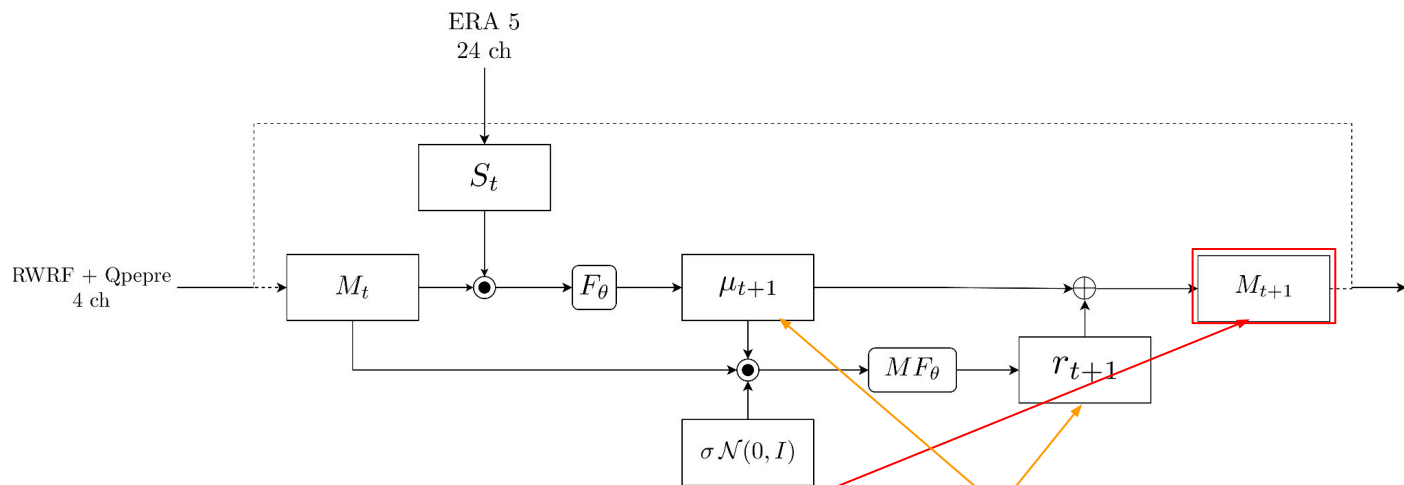
# Method: MeanFlow



$$\mathcal{L}_{\text{MF}} = \mathbb{E} \left[ \text{sg}(w) \left\| u_{\theta}(z_r, r, t, c_t) - \text{sg}(u_{\text{tgt}}) \right\|_{2, \beta}^2 \right] + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}}, \quad u_{\text{tgt}} = v + (t - r)(v \partial_z u_{\theta} + \partial_r u_{\theta}),$$

- Stop gradient (sg) keeps JVP first order, keeping the training stable.
- $\mathcal{L}_{\text{spec}}$ : Matches the precipitation power spectrum so rain stays sharp, not blurry.

# Method: Next hour prediction



$$p(M_{t+1} | M_t, S_t) = r_{t+1} + \mu_{t+1}$$

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# Experiments: Setup

- All models are trained for ~2 Million samples.
- Training dataset: 08/01/2019 - 12/31/2021
- Validation dataset: 01/01/2022 - 12/31/2022
- Ensembles: 5 members
- Sequences: 10 initial times
- Rollouts: 6 hours,  $dt = 1$  hour
- All scores are denormalised to physical units ( $\text{mm h}^{-1}$ , K,  $\text{ms}^{-1}$ )
- Legacy EDM is the StormCast that was trained on a slightly bigger domain (mismatches qpepre's domain)

# Experiments: Metrics

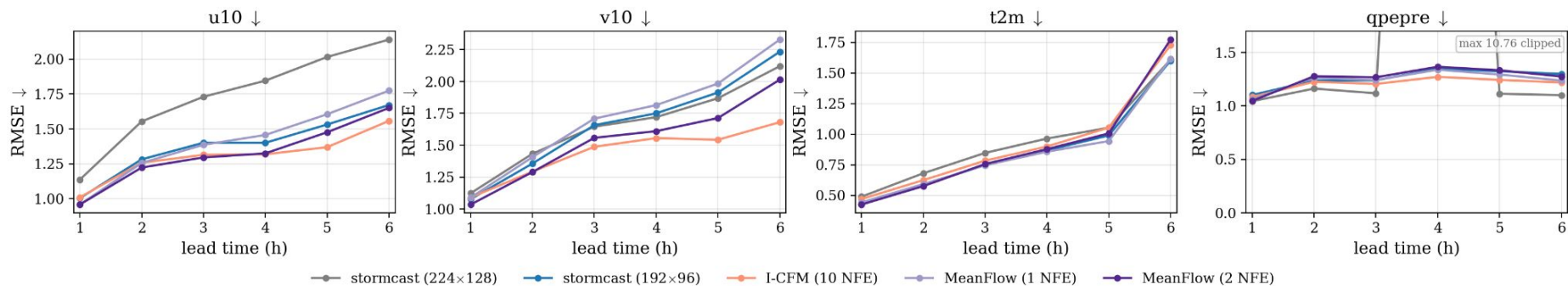
| Metric                      | Formula                                                                   | Meaning                                                                  |
|-----------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------|
| RMSE ↓                      | $\sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{x}_i - x_i)^2}$                     | Mean per-pixel error in physical units; reported per channel.            |
| CRPS ↓                      | $\frac{1}{K} \sum_k  x_k - y  - \frac{1}{2K(K-1)} \sum_{j,k}  x_j - x_k $ | Probabilistic MAE: rewards accuracy and a calibrated ensemble spread.    |
| CSI ↑                       | $\frac{TP}{TP + FP + FN}$                                                 | Hit fraction among forecast-or-observed rain pixels; ignores dry pixels. |
| HSS ↑                       | $\frac{TP + TN - E}{N - E}$                                               | Categorical skill above random chance $E$ ( $1 = \text{perfect}$ ).      |
| FAR ↓                       | $\frac{FP}{TP + FP}$                                                      | Fraction of forecast rain pixels that stayed dry.                        |
| FSS ↑                       | $1 - \frac{(P_f - P_o)^2}{P_f^2 + P_o^2}$                                 | Neighbourhood rain-fraction overlap; tolerant to small displacement.     |
| <i>Evaluation protocols</i> |                                                                           |                                                                          |
| Rollout                     | $\hat{x}_{t+k} = \mathcal{M}(\hat{x}_{t+k-1})$                            | Autoregressive feedback; tracks error growth vs lead time (1–6 h).       |
| Ensemble                    | $\{x_k\}_{k=1}^K, x_k = \mu + r_\theta(c, \epsilon_k)$                    | $K$ stochastic samples: mean is the forecast, spread the uncertainty.    |

# Experiments: Overview

| Method               | Time/seq (s) ↓ | Speedup vs EDM ↑ | CRPS ↓        | RMSE u10 ↓    | RMSE v10 ↓    | RMSE t2m ↓    | RMSE qpepre ↓ |
|----------------------|----------------|------------------|---------------|---------------|---------------|---------------|---------------|
| Legacy EDM (224×128) | 19.6630        | —                | 0.6348        | 1.6912        | 1.6063        | 0.8845        | 1.4827        |
| EDM diffusion        | 11.2300        | 1.0×             | 0.5537        | 1.3513        | 1.6082        | <b>0.8023</b> | 1.0292        |
| FlowCast ( $K=10$ )  | 3.0680         | 3.7×             | <b>0.5518</b> | <b>1.2838</b> | <b>1.4085</b> | 0.8498        | <b>0.9590</b> |
| MeanFlow ( $K=1$ )   | <b>0.9310</b>  | <b>12.1×</b>     | 0.6710        | 1.3767        | 1.6915        | 0.8087        | 0.9834        |
| MeanFlow ( $K=2$ )   | 1.1640         | 9.6×             | 0.6093        | 1.2986        | 1.4887        | 0.8472        | 1.0054        |

# Experiments: RMSE Rollout per hour

Per-channel RMSE vs lead time (main experiment) ↓





# Experiments: Thresholds (3 km, grid)

| Neighbourhood kernel: 15 km |        |               |               |               |               |               |
|-----------------------------|--------|---------------|---------------|---------------|---------------|---------------|
| Method                      | Metric | ≥ 0.1 mm/h    | ≥ 1.0 mm/h    | ≥ 5.0 mm/h    | ≥ 10.0 mm/h   | ≥ 16.0 mm/h   |
| stormcast (192×96)          | csi ↑  | 0.6498        | <b>0.4629</b> | 0.1570        | 0.0279        | 0.0134        |
| I-CFM                       | csi ↑  | 0.6598        | 0.4251        | 0.1290        | 0.0307        | 0.0073        |
| MeanFlow (1 NFE)            | csi ↑  | 0.6346        | 0.4053        | 0.1347        | 0.0414        | 0.0057        |
| MeanFlow (2 NFE)            | csi ↑  | <b>0.6807</b> | 0.4284        | <b>0.1657</b> | <b>0.0529</b> | <b>0.0172</b> |
| stormcast (192×96)          | far ↓  | 0.2903        | 0.4293        | 0.4462        | 0.5426        | <b>0.0000</b> |
| I-CFM                       | far ↓  | 0.2043        | <b>0.3223</b> | <b>0.3449</b> | <b>0.4428</b> | 0.4375        |
| MeanFlow (1 NFE)            | far ↓  | <b>0.1658</b> | 0.3288        | 0.4250        | 0.5731        | 0.0000        |
| MeanFlow (2 NFE)            | far ↓  | 0.1803        | 0.3594        | 0.4548        | 0.6015        | 0.2732        |
| stormcast (192×96)          | pod ↑  | <b>0.8850</b> | <b>0.7101</b> | 0.1797        | 0.0288        | 0.0134        |
| I-CFM                       | pod ↑  | 0.7943        | 0.5329        | 0.1384        | 0.0315        | 0.0073        |
| MeanFlow (1 NFE)            | pod ↑  | 0.7262        | 0.5056        | 0.1496        | 0.0438        | 0.0057        |
| MeanFlow (2 NFE)            | pod ↑  | 0.8006        | 0.5640        | <b>0.1922</b> | <b>0.0575</b> | <b>0.0174</b> |
| stormcast (192×96)          | hss ↑  | 0.6092        | <b>0.5236</b> | 0.2520        | 0.0519        | 0.0262        |
| I-CFM                       | hss ↑  | 0.6519        | 0.5069        | 0.2140        | 0.0575        | 0.0143        |
| MeanFlow (1 NFE)            | hss ↑  | 0.6376        | 0.4853        | 0.2204        | 0.0759        | 0.0113        |
| MeanFlow (2 NFE)            | hss ↑  | <b>0.6799</b> | 0.5038        | <b>0.2640</b> | <b>0.0958</b> | <b>0.0336</b> |
| stormcast (192×96)          | fss ↑  | 0.7794        | 0.5973        | 0.2668        | 0.0281        | 0.0076        |
| I-CFM                       | fss ↑  | 0.8046        | <b>0.6290</b> | 0.2262        | 0.0148        | 0.0038        |
| MeanFlow (1 NFE)            | fss ↑  | 0.8037        | 0.6153        | 0.2815        | 0.0416        | 0.0017        |
| MeanFlow (2 NFE)            | fss ↑  | <b>0.8079</b> | 0.6016        | <b>0.3094</b> | <b>0.0553</b> | <b>0.0264</b> |

# Experiments: Thresholds (15 km)

| Neighbourhood kernel: 15 km |                  |                 |                 |                 |                  |                  |
|-----------------------------|------------------|-----------------|-----------------|-----------------|------------------|------------------|
| Method                      | Metric           | $\geq 0.1$ mm/h | $\geq 1.0$ mm/h | $\geq 5.0$ mm/h | $\geq 10.0$ mm/h | $\geq 16.0$ mm/h |
| stormcast (192×96)          | csi $\uparrow$   | 0.6498          | <b>0.4629</b>   | 0.1570          | 0.0279           | 0.0134           |
| I-CFM                       | csi $\uparrow$   | 0.6598          | 0.4251          | 0.1290          | 0.0307           | 0.0073           |
| MeanFlow (1 NFE)            | csi $\uparrow$   | 0.6346          | 0.4053          | 0.1347          | 0.0414           | 0.0057           |
| MeanFlow (2 NFE)            | csi $\uparrow$   | <b>0.6807</b>   | 0.4284          | <b>0.1657</b>   | <b>0.0529</b>    | <b>0.0172</b>    |
| stormcast (192×96)          | far $\downarrow$ | 0.2903          | 0.4293          | 0.4462          | 0.5426           | <b>0.0000</b>    |
| I-CFM                       | far $\downarrow$ | 0.2043          | <b>0.3223</b>   | <b>0.3449</b>   | <b>0.4428</b>    | 0.4375           |
| MeanFlow (1 NFE)            | far $\downarrow$ | <b>0.1658</b>   | 0.3288          | 0.4250          | 0.5731           | 0.0000           |
| MeanFlow (2 NFE)            | far $\downarrow$ | 0.1803          | 0.3594          | 0.4548          | 0.6015           | 0.2732           |
| stormcast (192×96)          | pod $\uparrow$   | <b>0.8850</b>   | <b>0.7101</b>   | 0.1797          | 0.0288           | 0.0134           |
| I-CFM                       | pod $\uparrow$   | 0.7943          | 0.5329          | 0.1384          | 0.0315           | 0.0073           |
| MeanFlow (1 NFE)            | pod $\uparrow$   | 0.7262          | 0.5056          | 0.1496          | 0.0438           | 0.0057           |
| MeanFlow (2 NFE)            | pod $\uparrow$   | 0.8006          | 0.5640          | <b>0.1922</b>   | <b>0.0575</b>    | <b>0.0174</b>    |
| stormcast (192×96)          | hss $\uparrow$   | 0.6092          | <b>0.5236</b>   | 0.2520          | 0.0519           | 0.0262           |
| I-CFM                       | hss $\uparrow$   | 0.6519          | 0.5069          | 0.2140          | 0.0575           | 0.0143           |
| MeanFlow (1 NFE)            | hss $\uparrow$   | 0.6376          | 0.4853          | 0.2204          | 0.0759           | 0.0113           |
| MeanFlow (2 NFE)            | hss $\uparrow$   | <b>0.6799</b>   | 0.5038          | <b>0.2640</b>   | <b>0.0958</b>    | <b>0.0336</b>    |
| stormcast (192×96)          | fss $\uparrow$   | 0.7794          | 0.5973          | 0.2668          | 0.0281           | 0.0076           |
| I-CFM                       | fss $\uparrow$   | 0.8046          | <b>0.6290</b>   | 0.2262          | 0.0148           | 0.0038           |
| MeanFlow (1 NFE)            | fss $\uparrow$   | 0.8037          | 0.6153          | 0.2815          | 0.0416           | 0.0017           |
| MeanFlow (2 NFE)            | fss $\uparrow$   | <b>0.8079</b>   | 0.6016          | <b>0.3094</b>   | <b>0.0553</b>    | <b>0.0264</b>    |

# Experiments: Thresholds (27 km)

| Neighbourhood kernel: 27 km |                  |                 |                 |                 |                  |                  |
|-----------------------------|------------------|-----------------|-----------------|-----------------|------------------|------------------|
| Method                      | Metric           | $\geq 0.1$ mm/h | $\geq 1.0$ mm/h | $\geq 5.0$ mm/h | $\geq 10.0$ mm/h | $\geq 16.0$ mm/h |
| stormcast (192×96)          | csi $\uparrow$   | 0.6840          | <b>0.5479</b>   | <b>0.2014</b>   | 0.0454           | 0.0158           |
| I-CFM                       | csi $\uparrow$   | 0.7009          | 0.4933          | 0.1577          | 0.0537           | 0.0139           |
| MeanFlow (1 NFE)            | csi $\uparrow$   | 0.6594          | 0.4429          | 0.1574          | 0.0633           | 0.0085           |
| MeanFlow (2 NFE)            | csi $\uparrow$   | <b>0.7175</b>   | 0.4770          | 0.1830          | <b>0.0802</b>    | <b>0.0176</b>    |
| stormcast (192×96)          | far $\downarrow$ | 0.2629          | 0.3390          | 0.3603          | 0.3518           | <b>0.0000</b>    |
| I-CFM                       | far $\downarrow$ | 0.1776          | 0.2497          | <b>0.2764</b>   | <b>0.2844</b>    | 0.2390           |
| MeanFlow (1 NFE)            | far $\downarrow$ | <b>0.1291</b>   | <b>0.2391</b>   | 0.3311          | 0.3927           | 0.0000           |
| MeanFlow (2 NFE)            | far $\downarrow$ | 0.1389          | 0.2615          | 0.3984          | 0.4320           | 0.3402           |
| stormcast (192×96)          | pod $\uparrow$   | <b>0.9047</b>   | <b>0.7620</b>   | <b>0.2272</b>   | 0.0466           | 0.0158           |
| I-CFM                       | pod $\uparrow$   | 0.8259          | 0.5901          | 0.1678          | 0.0549           | 0.0140           |
| MeanFlow (1 NFE)            | pod $\uparrow$   | 0.7308          | 0.5145          | 0.1707          | 0.0660           | 0.0085           |
| MeanFlow (2 NFE)            | pod $\uparrow$   | 0.8114          | 0.5739          | 0.2082          | <b>0.0854</b>    | <b>0.0178</b>    |
| stormcast (192×96)          | hss $\uparrow$   | 0.5805          | <b>0.5796</b>   | <b>0.3029</b>   | 0.0821           | 0.0305           |
| I-CFM                       | hss $\uparrow$   | 0.6465          | 0.5470          | 0.2473          | 0.0968           | 0.0268           |
| MeanFlow (1 NFE)            | hss $\uparrow$   | 0.6215          | 0.4986          | 0.2447          | 0.1122           | 0.0166           |
| MeanFlow (2 NFE)            | hss $\uparrow$   | <b>0.6796</b>   | 0.5283          | 0.2763          | <b>0.1395</b>    | <b>0.0337</b>    |
| stormcast (192×96)          | fss $\uparrow$   | 0.8088          | 0.6356          | 0.3202          | 0.0383           | 0.0065           |
| I-CFM                       | fss $\uparrow$   | 0.8357          | <b>0.6768</b>   | 0.2673          | 0.0177           | 0.0040           |
| MeanFlow (1 NFE)            | fss $\uparrow$   | 0.8325          | 0.6551          | 0.3300          | 0.0510           | 0.0014           |
| MeanFlow (2 NFE)            | fss $\uparrow$   | <b>0.8361</b>   | 0.6344          | <b>0.3612</b>   | <b>0.0619</b>    | <b>0.0228</b>    |

# Outline

- Introduction
- Related work
- Dataset
- Method
- Experiments
- Discussion
- Conclusion

# Discussion

- At the cost of 1-2 NFEs, MeanFlow is able to reach the same forecasting quality as StormCast's EDM, reducing inference cost in terms of time by ~10 times.
- This can be directly applied by similar use case that computational usage is a concern. (e.g. lower time/spatial resolution)
- Failure modes:
  - Heavy rain still under-detection
  - Under-dispersion at  $K=1$ ,  $K=2$  is recommended

# Outline

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# Conclusion

- Contribution:
  - First application of the MeanFlow objective to Taiwan's convection-allowing hourly forecasting model.
  - Reduces inference time by  $\sim 10$  times while achieving similar/better forecasting quality.
- Future work:
  - Apply to application of lower time/space resolution.
  - Restore low-NFE rollout spread.
  - Tackle heavy-rain under-detection.

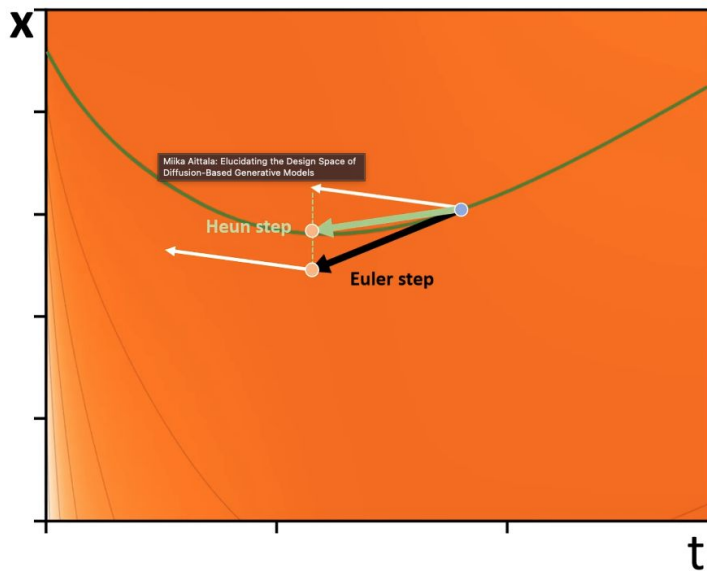
# EDM

- Noise scheduling: Exactly how much noise is injected or removed at any given moment.
- Preconditioning: How the inputs and outputs are **scaled** so the neural network doesn't mathematically choke on extreme values.



# EDM

- The Solver:



## Higher-order solvers

- Clever sub-steps  $\rightarrow$   
higher accuracy,  
higher cost
- Empirically, 2<sup>nd</sup> order **Heun method** strikes best balance

# StormCast (Nvidia, 2024), collaboration with NCDR

